

Midterm Exam 1

ECON 320 – Introduction to Mathematical Economics

October 01, 2025

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Instructions:

- You are allowed **75 minutes** to complete the exam.
 - Show all work. Unsupported answers receive little credit.
 - Unless stated otherwise, all variables are real and differentiable as needed.
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1. **(7 points)** Let $U = \mathbb{R}$. Define

$$A = \{x \in \mathbb{R} : x^2 - 7x + 12 \geq 0\}, \quad B = \{x \in \mathbb{R} : |x - 4| \leq 3\}, \quad C = \{x \in \mathbb{R} : 1 < x < 5\}.$$

- (a) Write A , B , and C explicitly as unions of intervals. (3)
- (b) Compute $A \cap B$, $A \cup C$, and $B \setminus C$. (2)
- (c) Decide whether $(A \cup B) \setminus C = (A \setminus C) \cup (B \setminus C)$. (2)

2. **(8 points)** Let

$$u(x, y) = x^2y - y^3 + \ln(1 + xy).$$

Suppose

$$x = r^2, \quad y = s^3, \quad r > 0, \quad s > 0.$$

- (a) Compute $\partial u / \partial x$ and $\partial u / \partial y$. (3)
- (b) Using the chain rule, compute $\partial u / \partial r$ and $\partial u / \partial s$ in terms of r, s . (5)

3. **(8 points)** Let $F(x, y) = \ln(y + 2x) - x^2 + 3y - 10 = 0$ define y as a function of x near $x = 1$.

- (a) Find y such that $F(1, y) = 0$ and verify the implicit function conditions at that point. (3)
- (b) Compute $\frac{dy}{dx}$ at that point. (3)
- (c) Approximate the change in y when x increases from 1 to 1.04 using the total differential. (2)

4. **(10 points)** Consider the transformation $(u, v, w) \mapsto (x, y, z)$:

$$x = u^2 + 2vw, \quad y = v^2 + 2uw, \quad z = w^2 + 2uv.$$

- (a) Compute the Jacobian matrix $J = \partial(x, y, z)/\partial(u, v, w)$. (5)
 (b) Evaluate $\det(J)$ at $(u, v, w) = (1, -2, 3)$. Is the mapping locally invertible there? (5)

5. **(17 points)** A consumer maximizes utility

$$U(x_1, x_2, \ell) = \ln(x_1) + \ln(x_2) + \beta \ln(\ell),$$

where x_1 and x_2 are quantities of two goods and ℓ is leisure. The consumer faces the budget/time constraint:

$$p_1x_1 + p_2x_2 + w\ell = I,$$

where p_1, p_2 are prices, w is the wage (price of leisure), and I is income. The first-order conditions are:

$$\begin{cases} F_1 = p_1x_1 + p_2x_2 + w\ell - I = 0, \\ F_2 = \frac{1}{x_1} - \lambda p_1 = 0, \\ F_3 = \frac{1}{x_2} - \lambda p_2 = 0. \end{cases}$$

- (a) Write the Jacobian $J = \partial(F_1, F_2, F_3)/\partial(x_1, x_2, \ell)$. (5)
 (b) Suppose $p_1 = 2$, $p_2 = 1$, $w = 1$, $\beta = 1$, and $I = 12$. At the bundle $(x_1, x_2, \ell) = (2, 4, 6)$, compute J and $\det(J)$. (7)
 (c) Using Cramer's Rule logic, determine the signs of $\frac{\partial x_1}{\partial I}$, $\frac{\partial x_2}{\partial I}$, $\frac{\partial \ell}{\partial I}$. Provide an economic interpretation. (5)